

POWER: USB-C, ESD, Li-Po charger, power path, 3.3 V LDO

MCU: ESP32-S3-WROOM-1 (N16R8), reset/boot, status LED, battery sense, UART pads

CAMERA: DVP camera (OV2640/OV5640 24-pin FPC) + camera LDOS

AUDIO: I2S MEMS microphone, I2S class-D amp, bone-conduction transducer

GPIO map: IO4 VBAT Sense | IO5/6/7 MIC SCK/WS/SD (I2S0) | IO15/16/17 AMP BCLK/LRCLK/DIN (I2S1) | IO18 AMP_SD
IO8 status LED | IO19/20 USB D-/D+ | IO9-14,21,47 CAM DO-D7 | IO48 PCLK | IO38 VSYNC | IO39 HREF | IO40 XCLK
IO41 SIOD | IO42 SIOC | IO2 PWDN | IO1 RESET | IO0 BOOT button | IO43/44 UART test pads
Do not use IO35/36/37 (octal PSRAM on N16R8). IO3/IO45/IO46 are strapping pins and left unconnected.
Charge current = 1000 V / R₄ = 196 mA (R₄ = 5.1k). Change R₄ to suit the battery (<= 1C).

Power flags (tell ERC these nets are driven)

Generated from scripts/design.py – edit that file and regenerate
ESP32–S3, DVP camera, I2S mic + amp, bone transducer, Li–Po
Trendier
Sheet: /
File: ai_glasses.kicad_sch
Title: AI Glasses — Temple PCB
Size: A2 Date: 2026–09–30 Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dsg-1build4 Id: 1/1

POWER: USB-C, ESD, Li-Po charger, power path, 3.3 V LDO

MCU: ESP32-S3-WROOM-1 (N16R8), reset/boot, status LED, battery sense, UART pads

CAMERA: DVP camera (OV2640/OV5640 24-pin FPC) + camera LDOS

AUDIO: I2S MEMS microphone, I2S class-D amp, bone-conduction transducer

GPIO map: IO4 VBAT_SENSE | IO5/6/7 MIC SCK/WS/SD (I2S0) | IO15/16/17 AMP BCLK/LRCLK/DIN (I2S1) | IO18 AMP_SD
IO8 status LED | IO19/20 USB D-/D+ | IO9-14,21,47 CAM DO-D7 | IO48 PCLK | IO38 VSYNC | IO39 HREF | IO40 XCLK
IO41 SIOD | IO42 SIOD | IO2 PWDN | IO1 RESET | IO0 BOOT button | IO43/44 UART test pads

Do not use IO35/36/37 (octal PSRAM on N16R8). IO3/IO45/IO46 are strapping pins and left unconnected.

Charge current = 1000 V / R4 = 196 mA (R4 = 5.1k). Change R4 to suit the battery (<= 1C).

Power flags (tell ERC these nets are driven)

Generated from scripts/design.py – edit that file and regenerate			
ESP32-S3, DVP camera, I2S mic + amp, bone transducer, Li-Po			
Trendier			
Sheet: /			
File: ai_glasses.kicad_sch			
Title: AI Glasses – Temple PCB			
Size: A2	Date: 2026-09-30	Rev: 1.0	
KiCad E.D.A.	kicad-cli 7.0.11+dsg-1build4	Id: 1/1	

POWER: USB-C, ESD, Li-Po charger, power path, 3.3 V LDO

MCU: ESP32-S3-WROOM-1 (N16R8), reset/boot, status LED, battery sense, UART pads

CAMERA: DVP camera (OV2640/OV5640 24-pin FPC) + camera LDOs

AUDIO: I2S MEMS microphone, I2S class-D amp, bone-conduction transducer

GPIO map: IO4 VBAT_SENSE | IO5/6/7 MIC SCK/WS/SD (I2S0) | IO15/16/17 AMP BCLK/LRCLK/DIN (I2S1) | IO18 AMP_SD
IO8 status LED | IO19/20 USB D-/D+ | IO9-14,21,47 CAM DO-D7 | IO48 PCLK | IO38 VSYNC | IO39 HREF | IO40 XCLK
IO41 SIOD | IO42 SIOC | IO2 PWDN | IO1 RESET | IO0 BOOT button | IO43/44 UART test pads

Do not use IO35/36/37 (octal PSRAM on N16R8). IO3/IO45/IO46 are strapping pins and left unconnected.
Charge current = 1000 V / R4 = 196 mA (R4 = 5.1k). Change R4 to suit the battery (<= 1C).

Power flags (tell ERC these nets are driven)

Generated from scripts/design.py – edit that file and regenerate
ESP32-S3, DVP camera, I2S mic + amp, bone transducer, Li-Po
Trendier
Sheet: /
File: ai_glasses.kicad_sch
Title: AI Glasses – Temple PCB
Size: A2 Date: 2026-09-30 Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dsg-1build4 Id: 1/1

POWER: USB-C, ESD, Li-Po charger, power path, 3.3 V LDO

MCU: ESP32-S3-WROOM-1 (N16R8), reset/boot, status LED, battery sense, UART pads

CAMERA: DVP camera (OV2640/OV5640 24-pin FPC) + camera LDOs

AUDIO: I2S MEMS microphone, I2S class-D amp, bone-conduction transducer

GPIO map: IO4 VBAT_SENSE | IO5/6/7 MIC SCK/WS/SD (I2S0) | IO15/16/17 AMP BCLK/LRCLK/DIN (I2S1) | IO18 AMP_SD
IO8 status LED | IO19/20 USB D-/D+ | IO9-14,21,47 CAM DO-D7 | IO48 PCLK | IO38 VSYNC | IO39 HREF | IO40 XCLK
IO41 SIOD | IO42 SIOC | IO2 PWDN | IO1 RESET | IO0 BOOT button | IO43/44 UART test pads
Do not use IO35/36/37 (octal PSRAM on N16R8). IO3/IO45/IO46 are strapping pins and left unconnected.
Charge current = 1000 V / R4 = 196 mA (R4 = 5.1k). Change R4 to suit the battery (<= 1C).

Power flags (tell ERC these nets are driven)

VBUS	PWR_FLAG	GND	PWR_FLAG
VBAT	PWR_FLAG	VSYS	PWR_FLAG

Generated from scripts/design.py – edit that file and regenerate
ESP32-S3, DVP camera, I2S mic + amp, bone transducer, Li-Po
Trendier
Sheet: /
File: ai_glasses.kicad_sch
Title: AI Glasses – Temple PCB
Size: A2 Date: 2026-09-30 Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dsg-1build4 Id: 1/1

POWER: USB-C, ESD, Li-Po charger, power path, 3.3 V LDO

MCU: ESP32-S3-WROOM-1 (N16R8), reset/boot, status LED, battery sense, UART pads

CAMERA: DVP camera (OV2640/OV5640 24-pin FPC) + camera LDOs

AUDIO: I2S MEMS microphone, I2S class-D amp, bone-conduction transducer

GPIO map: IO4 VBAT_SENSE | IO5/6/7 MIC SCK/WS/SD (I2S0) | IO15/16/17 AMP BCLK/LRCLK/DIN (I2S1) | IO18 AMP_SD IO8 status LED | IO19/20 USB D-/D+ | IO9-14,21,47 CAM D0-D7 | IO48 PCLK | IO38 VSYNC | IO39 HREF | IO40 XCLK IO41 SIOD | IO42 SIOC | IO2 PWDN | IO1 RESET | IO0 BOOT button | IO43/44 UART test pads
Do not use IO35/36/37 (octal PSRAM on N16R8). IO3/IO45/IO46 are strapping pins and left unconnected.
Charge current = 1000 V / R4 = 196 mA (R4 = 5.1k). Change R4 to suit the battery (<= 1C).

Power flags (tell ERC these nets are driven)

Generated from scripts/design.py – edit that file and regenerate
ESP32-S3, DVP camera, I2S mic + amp, bone transducer, Li-Po
Trendier
Sheet: /
File: ai_glasses.kicad_sch
Title: AI Glasses – Temple PCB
Size: A2 | Date: 2026-09-30 | Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1

POWER: USB-C, ESD, Li-Po charger, power path, 3.3 V LDO

MCU: ESP32-S3-WROOM-1 (N16R8), reset/boot, status LED, battery sense, UART pads

CAMERA: DVP camera (OV2640/OV5640 24-pin FPC) + camera LDOs

AUDIO: I2S MEMS microphone, I2S class-D amp, bone-conduction transducer

GPIO map: IO4 VBAT_SENSE | IO5/6/7 MIC SCK/WS/SD (I2S0) | IO15/16/17 AMP BCLK/LRCLK/DIN (I2S1) | IO18 AMP_SD
IO8 status LED | IO19/20 USB D-/D+ | IO9-14,21,47 CAM DO-D7 | IO48 PCLK | IO38 VSYNC | IO39 HREF | IO40 XCLK
IO41 SIOD | IO42 SIOC | IO2 PWDN | IO1 RESET | IO0 BOOT button | IO43/44 UART test pads
Do not use IO35/36/37 (octal PSRAM on N16R8). IO3/IO45/IO46 are strapping pins and left unconnected.
Charge current = 1000 V / R4 = 196 mA (R4 = 5.1k). Change R4 to suit the battery (<= 1C).

Power flags (tell ERC these nets are driven)

VBUS	PWR_FLAG	GND	PWR_FLAG
VBAT	PWR_FLAG	VSYS	PWR_FLAG

Generated from scripts/design.py – edit that file and regenerate
ESP32-S3, DVP camera, I2S mic + amp, bone transducer, Li-Po
Trendier
Sheet: /
File: ai_glasses.kicad_sch
Title: AI Glasses – Temple PCB
Size: A2 Date: 2026-09-30 Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dsg-1build4 Id: 1/1